Brain Tumor Classification using CNN

Comprehensive Project Presentation & External Viva Examination Guide

Architecture: DenseNet201 with Custom Transfer Learning Classifier

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1. Executive Project Summary

This project provides an automated, computer-aided diagnostic system to classify Magnetic Resonance Imaging (MRI) brain scans into four categories: three malignant/benign brain tumor types (**Glioma, Meningioma, Pituitary**) and a **No Tumor** healthy control group. By leveraging **Transfer Learning** with the **DenseNet201** architecture, the model achieves high diagnostic accuracy, mitigating the scarcity of specialized radiologists and providing rapid, objective second opinions.

2. 4-Member Team Role Division & Responsibilities

Member	Presentation Topic	Key Responsibilities & Deliverables
Member 1	Introduction & Problem Statement	What the project is, motivation, real-world clinical challenges, and project objectives.
Member 2	Dataset & Image Preprocessing	Dataset source, 4 tumor classes, image resizing (224x224), normalization, and data augmentation.
Member 3	Deep Learning Architecture	DenseNet201 backbone, feature reuse, custom classifier head (Flatten, Dense, Dropout, Softmax), loss function.
Member 4	Live Implementation & Conclusion	Demonstrating web app (Streamlit) and CLI, training curves analysis, metrics, and future scope.

3. Detailed Presentation Scripts (Member by Member)

Member 1: Project Overview & Motivation

Script to memorize & speak:

*"Good morning, respected teachers and external examiners. Our project is titled **Brain Tumor Classification using Deep Learning and CNN**. Brain tumors are among the most lethal neurological diseases. Every year, thousands of patients undergo MRI scans, but the manual analysis of these scans is time-consuming and subjective to human fatigue and inter-observer variability.*

In developing regions, the acute shortage of trained neuro-radiologists causes significant diagnostic delays. Our primary goal was to engineer an automated, high-accuracy classification pipeline capable of identifying specific tumor categories within seconds, providing clinicians with a dependable diagnostic second opinion."

Member 2: Dataset, Classes & Preprocessing Pipeline

Script to memorize & speak:

*"We utilized the benchmark Brain Tumor MRI dataset created by Sartaj Bhuvaji et al. The dataset comprises **3,264 MRI slices** divided into **2,870 training images** and **394 testing images** across four classes: **Glioma Tumor** (malignant brain tissue cancer), **Meningioma Tumor** (membrane-originating tumor), **Pituitary Tumor** (endocrine base tumor), and **No Tumor** (healthy scans).*

Our preprocessing pipeline implements three critical steps:
*1. **Resizing:** Normalized all scan resolutions to 224x224 pixels with 3 channels.*
*2. **Pixel Normalization:** Scaled pixel intensities by 1/255 into [0, 1] to accelerate gradient descent.*
*3. **Data Augmentation:** Applied real-time shear, zoom (20%), and horizontal flips to prevent overfitting and ensure rotational invariance."*

Member 3: Deep Learning Engine & DenseNet201

Script to memorize & speak:

*"Rather than training a CNN from random weights on a modest medical dataset, we implemented **Transfer Learning** using **DenseNet201** pre-trained on ImageNet. In DenseNet, each layer receives feature maps from all prior layers, which resolves vanishing gradients, encourages feature reuse, and drastically reduces required parameters.*

We froze the 201 base convolutional layers to preserve visual feature extraction, and added our custom top classifier:
*• **Flatten Layer:** Converts multidimensional feature maps into a 1D tensor.*
*• **Dense Layer (256 units, ReLU):** Learns high-level tumor discrimination patterns.*
*• **Dropout (0.3):** Randomly deactivates 30% of neurons to prevent co-adaptation.*
*• **Dense Layer (4 units, Softmax):** Outputs normalized probabilities summing to 1.0 across the 4 classes.*
*Trained using **Categorical Crossentropy loss** and **RMSprop optimizer** (LR = 0.00001)."*

Member 4: Live Demonstration, Results & Future Scope

Script to memorize & speak:

*"For clinical deployment, we developed an interactive **Streamlit Web Application** and command-line interface. When an MRI scan is uploaded, our pipeline processes the image and returns the predicted diagnosis, confidence score, and a complete probability breakdown across all tumor types in under one second.*

Our evaluation metrics demonstrate steady loss reduction and accuracy improvement, with model weights

> *checkpointed into an HDF5 binary file (`densenet101.h5`).*
> ***Future Enhancements:*** *We plan to integrate* ***Grad-CAM visual heatmaps*** *to visually highlight tumor boundaries for explainable AI, and expand the model to handle multi-sequence 3D volumetric MRI scans."*

4. Dataset Class Distribution & Statistics

Tumor Class	Clinical Origin & Characteristics	Training	Testing	Total
Glioma Tumor	Cancerous glial cells; infiltrative, requires aggressive treatment.	826	100	926
Meningioma	Arises from meninges membranes; mostly benign, slow growing.	822	115	937
Pituitary Tumor	Base skull adenoma affecting hormonal equilibrium.	827	74	901
No Tumor	Normal control brain scan without focal mass or edema.	395	105	500
Total Dataset	**Benchmark Dataset by Sartaj Bhuvaji et al.**	**2,870**	**394**	**3,264**

5. Complete Technology Stack

Technology Layer	Tool / Library	Version	Role in System
Programming Language	Python	3.11	Core development and neural computation pipeline.
Deep Learning Framework	TensorFlow	2.21	Backpropagation, tensor optimization, CPU computation.
High-Level Neural API	Keras	3.15	Model architecture, layers (Dense, Dropout, Flatten).
Transfer Learning Model	DenseNet201	ImageNet	201-layer deep feature extraction backbone.
Web Application UI	Streamlit	1.66	Interactive front-end dashboard for real-time MRI upload.
Image Processing	Pillow & Keras Preprocessing	13.0	Resizing, normalization, data augmentation generator.
Numerical & Science	NumPy & SciPy	2.4 / 1.17	Tensor shaping, probability array parsing, math operations.
Data Visualization	Matplotlib	3.11	Plotting training loss and validation accuracy curves.
Model Checkpoint Format	HDF5 (.h5)	Binary	Stores 42.4M trained weights in densenet101.h5 (268 MB).

6. Step-by-Step VS Code Demonstration Walkthrough

Follow these exact steps during your practical presentation:

Step 1: Open VS Code Terminal

Press **Ctrl + ~** in VS Code to display the PowerShell terminal.

Step 2: Launch the Interactive Web App

Execute: .\Brain-Tumar-classificaltion-using-CNN-main\.venv\Scripts\python.exe -m streamlit run app.py
Your browser will immediately open **http://localhost:8501**.

Step 3: Test Real MRI Scans Live

Under the **Sample MRI Scans** dropdown in the left sidebar, choose **Glioma Tumor** or **No Tumor**.
Show the examiner how the system instantly renders the predicted diagnosis, confidence percentage, and the probability breakdown across all 4 categories.

Step 4: Demonstrate Command-Line Prediction

To demonstrate backend scripting, run in terminal:
.\Brain-Tumar-classificaltion-using-CNN-main\.venv\Scripts\python.exe predict.py --image "Brain-Tumar-classificaltion-using-CNN-main\input\brain-tumor-classification-mri\Testing\no_tumor\image(10).jpg"

Step 5: Show Model Architecture Code

Open **train.py** and scroll to **build_model()** to explain the DenseNet201 base and custom Flatten -> Dense(256) -> Dropout(0.3) -> Dense(4, Softmax) head.

Step 6: Show Saved Artifacts

Show **training_curves.png** for the accuracy/loss progression, and **models/densenet101.h5** (268 MB binary weight file).

7. External Examiner Viva Q&A; Cheat Sheet

Q1: What is Transfer Learning and why did you use it?

Answer: *Transfer Learning reuses feature weights learned from a massive benchmark dataset (ImageNet with 1.4M images) as a foundation for a new, specialized task. This prevents overfitting on small medical datasets and drastically reduces training time.*

Q2: Why DenseNet201 instead of standard CNNs like VGG16 or ResNet?

Answer: *DenseNet connects each layer to every subsequent layer in a feed-forward manner. This ensures maximum feature reuse, mitigates the vanishing gradient problem, and requires fewer parameters than ResNet or VGG while achieving superior accuracy.*

Q3: Why Softmax activation for the output layer?

Answer: *Softmax normalizes the raw neural output logits into probability values between 0.0 and 1.0 that sum to exactly 1.0 (100%), which is mathematically required for mutually exclusive multi-class classification.*

Q4: What is the function of Dropout (0.3)?

Answer: *Dropout randomly turns off 30% of neuron connections during each training forward-pass. This prevents neurons from co-adapting and memorizing the training images, directly curbing overfitting.*

Q5: Why is the .h5 file not readable in a text editor?

Answer: *The .h5 (Hierarchical Data Format 5) file is a compiled binary array holding 42.4 million 32-bit floating-point numbers representing neural network weights and layer parameters. It is designed for high-speed numerical loading by TensorFlow, not text viewing.*

Q6: Why is the input resolution 224 x 224 x 3?

Answer: *DenseNet201 was originally trained on ImageNet images of size 224x224x3. Retaining this resolution allows us to directly utilize its pre-trained convolutional filters without re-engineering spatial dimensions.*

Q7: What is Categorical Crossentropy?

Answer: *It is the loss function used when true targets are one-hot encoded multi-class vectors. It measures the performance of a classification model whose output is a probability value between 0 and 1.*